

# Architecture

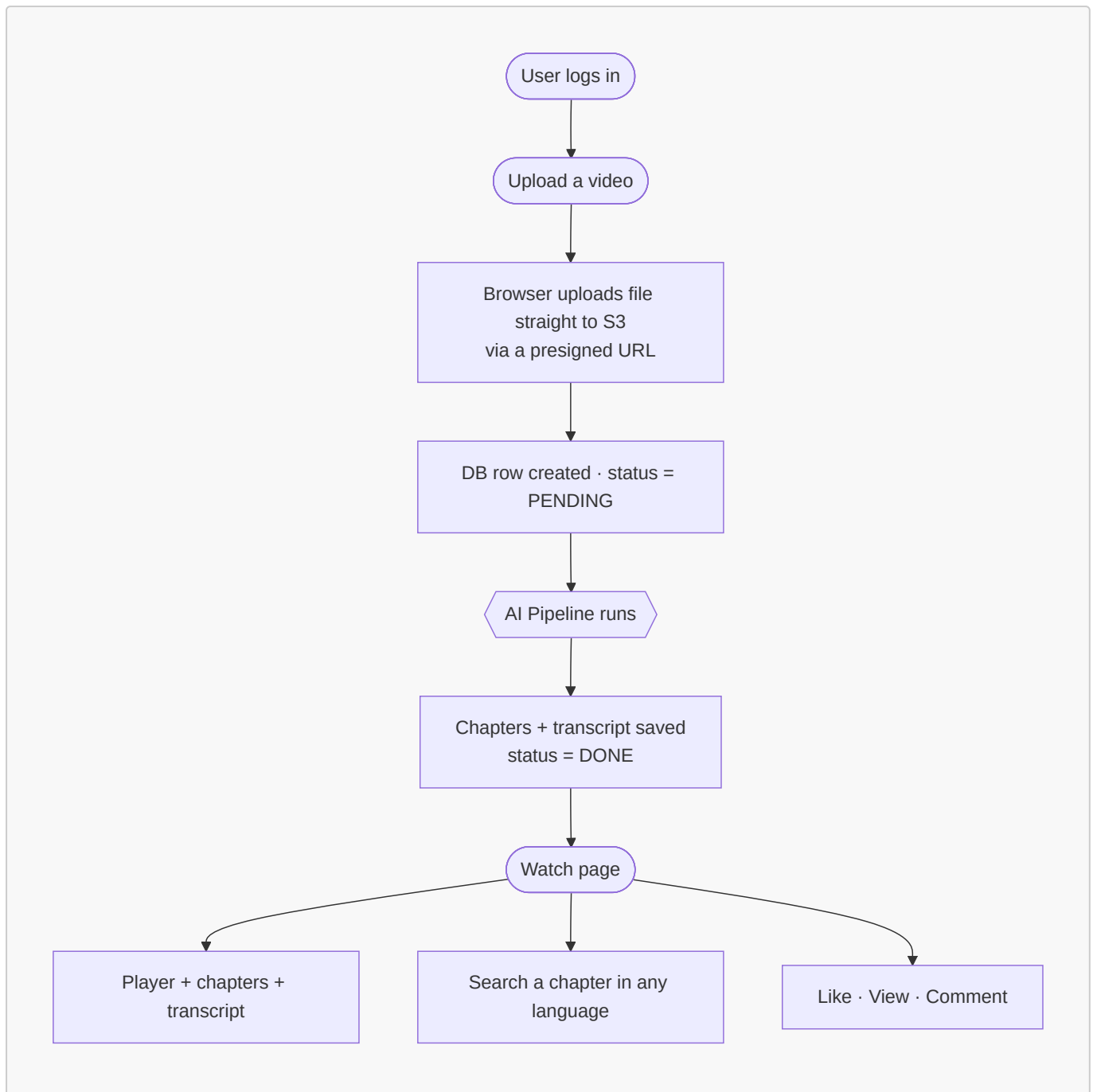
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## Stack

| Layer                     | Tech                                                    |
|---------------------------|---------------------------------------------------------|
| Frontend                  | Next.js 14 App Router · Tailwind CSS                    |
| Auth                      | NextAuth.js (email + bcrypt)                            |
| Database                  | Neon PostgreSQL via Prisma                              |
| File Storage              | AWS S3 (browser uploads directly via presigned URL)     |
| Transcription             | Groq Whisper ( <a href="#">whisper-large-v3-turbo</a> ) |
| Tagging / Search / Vision | OpenAI GPT-4o-mini · GPT-4o                             |
| Video Processing          | ffmpeg-static (audio · silence detect · thumbnails)     |

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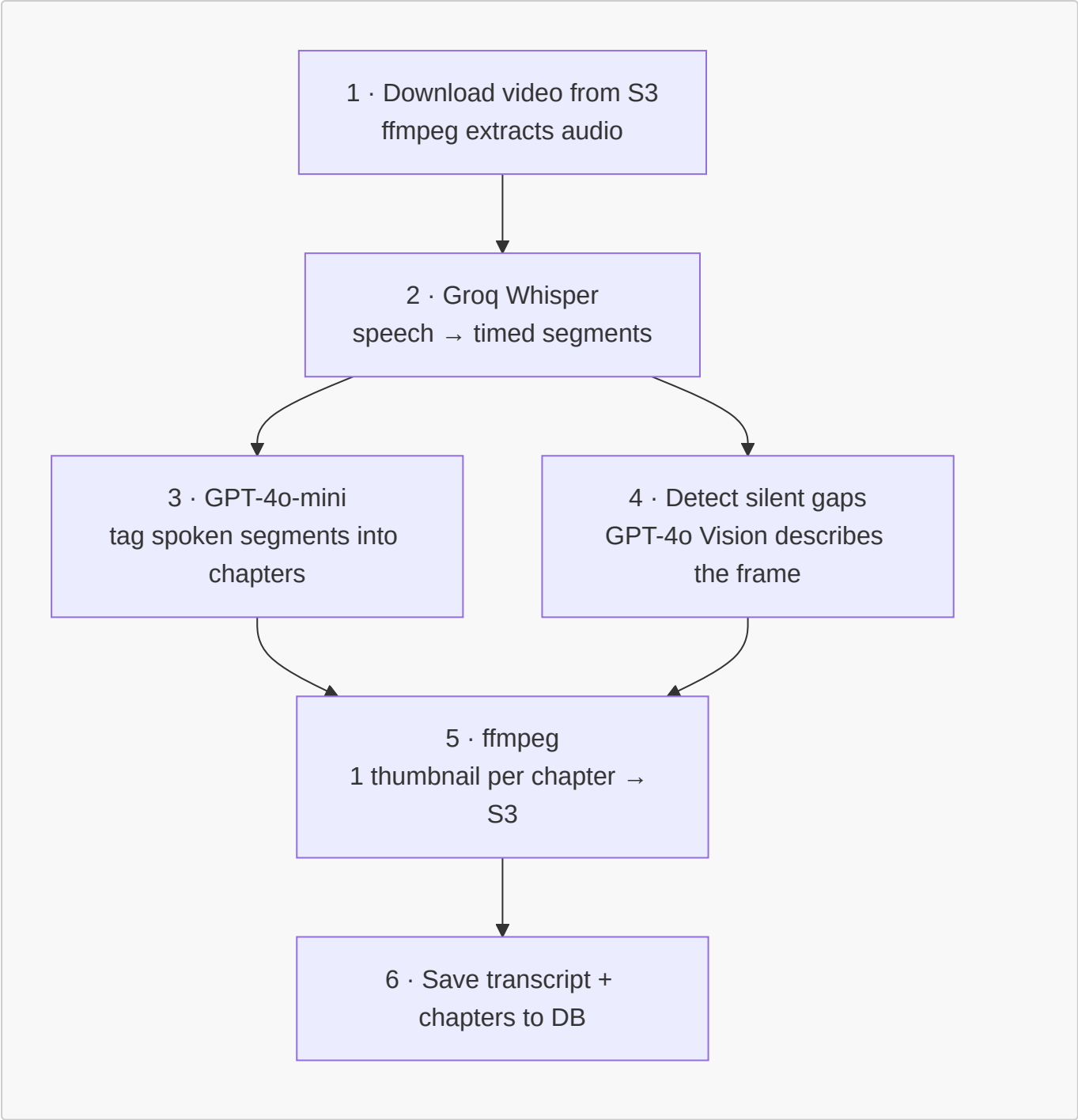
## System Flow (top → bottom)



While the pipeline runs, the page polls every 2s until status is **DONE**.

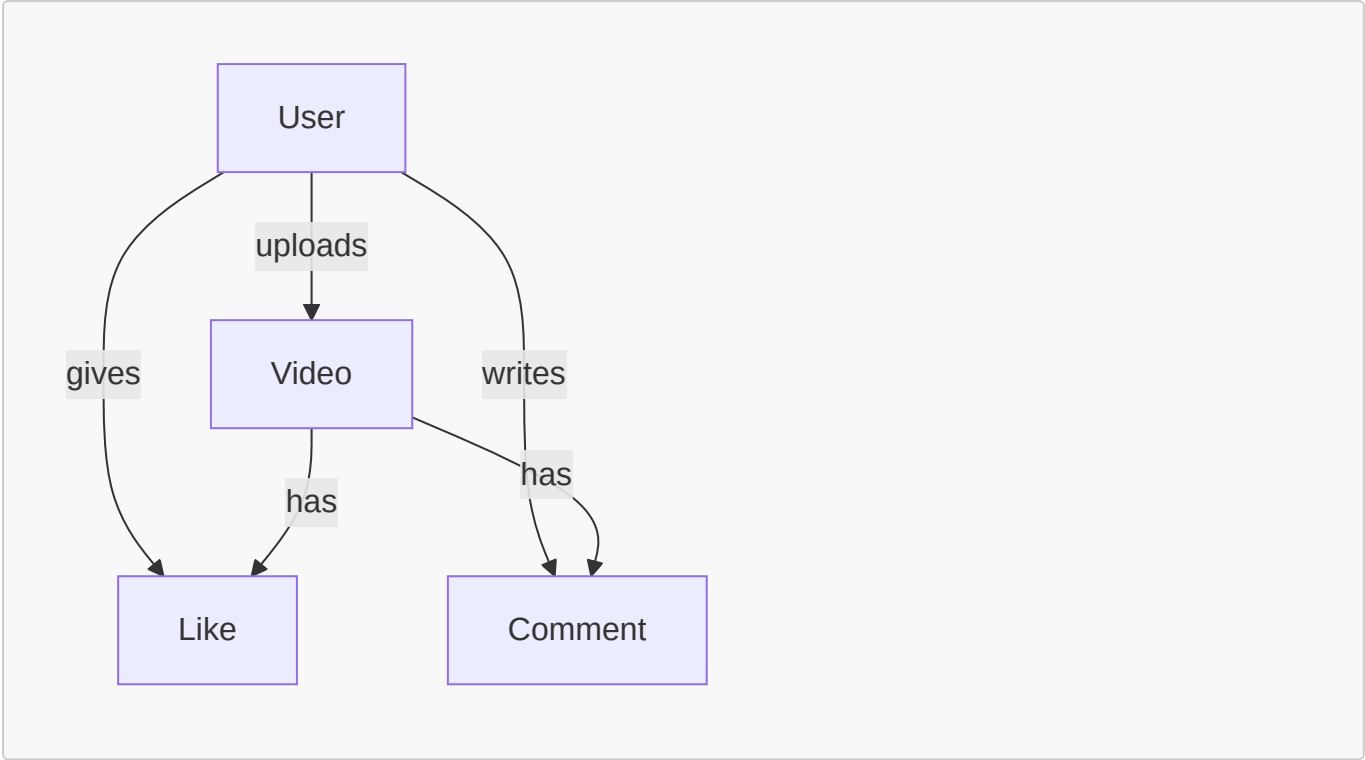
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## The AI Pipeline (top → bottom)



| Step | What it does                                                                                    |
|------|-------------------------------------------------------------------------------------------------|
| 1    | Pull the video from S3, extract a 16kHz mono audio track.                                       |
| 2    | Groq Whisper transcribes speech into timed segments (noise/hallucinations filtered).            |
| 3    | GPT-4o-mini picks the video's phases and tags each segment → <b>chapters</b> .                  |
| 4    | Silent stretches get a frame described by GPT-4o Vision → so wordless videos also get chapters. |
| 5    | ffmpeg grabs one thumbnail per chapter and uploads it to S3.                                    |
| 6    | Everything is saved to Postgres; status becomes <b>DONE</b> .                                   |

Data Model



`Video.transcriptStatus`: PENDING → PROCESSING → DONE (or FAILED). `transcriptSegments` = full timed transcript · `topicSegments` = the chapters (tag, time range, thumbnail).

## Key files

| File                                                          | Role                                             |
|---------------------------------------------------------------|--------------------------------------------------|
| <code>src/app/api/upload/route.ts</code>                      | Presigned S3 URL + create <code>Video</code> row |
| <code>src/app/api/videos/[id]/transcribe/route.ts</code>      | The full AI pipeline (steps 1–6)                 |
| <code>src/app/api/videos/[id]/transcript/route.ts</code>      | Status / transcript polling                      |
| <code>src/app/api/videos/[id]/search-chapter/route.ts</code>  | Multilingual chapter search                      |
| <code>src/lib/s3.ts</code>                                    | S3 helpers (presign · upload · download)         |
| <code>src/lib/auth.ts</code> · <code>src/lib/prisma.ts</code> | NextAuth config · Prisma client                  |